



ForgeGIS™

GPU-Accelerated Geospatial Primitives, Native to MCP

To our knowledge, the only GPU-accelerated geospatial library that exposes its full operation catalog to AI agents through a native Model Context Protocol server. 220 operations · 218 MCP-accessible · 62 MCP tools across two complementary surfaces. Every operation runs GPU-resident, numerically deterministic, and reachable from any MCP-capable agent — agents reach professional-grade GIS analysis through MCP without needing to know geospatial internals.

218 / 220

**operations
MCP-accessible**

two Java-only ops by design

62

**MCP tools
across two surfaces**

54 intent + 5 primitive + 3 escape

~800 ms

**warm-restart cold start
to first tools/list**

~3s on first-deploy, no OS cache

Two surfaces, one core

The MCP server exposes the catalog through two complementary access patterns. A *pipeline DSL* lets agents compose any of 133 operations into multi-step workflows the server executes GPU-resident, with intermediate results staying on the device. *54 curated intent tools* wrap common analytic tasks as typed single-call endpoints with strict input schemas. Same operations underneath — different access patterns for different agent workflows.

Deterministic by construction

The MCP layer is a parameter-marshaling shim over the same compute implementations the direct Java API exposes — not a parallel implementation. Numerical outputs depend on the operation and its parameters, not the invocation surface. Two operations involve randomness (*KMeans-Classify*, *RandomForest*) and use a fixed deterministic default seed under MCP invocation, so agent runs are reproducible without per-call seed configuration.

Built to be embedded

Pure Java, no JNI, no native dependencies. The shaded MCP server jar drops into any host environment with a JDK and a GPU driver — no CUDA-only runtime requirement, no system-level GDAL install, no platform-specific binaries. Communicates over JSON-RPC on stdio per the MCP specification. Agents that spawn MCP servers per session see approximately 800 ms median to first response on warm restart.

WHO THIS IS FOR

LLM agent platforms grounding spatial reasoning in real geometry

Agents that need to reason about where things are, what overlaps what, how far apart features sit, what a flood envelope looks like, or what's visible from a vantage point — and need the answer in milliseconds, not seconds. ForgeGIS exposes 218 operations through MCP without requiring the agent itself to understand projections, predicates, or pipeline composition.

Workflow and orchestration platforms

LangChain, AutoGen, MCP-native runners, and the broader class of platforms that compose tool calls into agent workflows. The dual-surface API — DSL for novel composition, intent tools for typed call shapes — lets the same library serve both "let the agent figure out the chain" and "give the agent a stable typed endpoint" workflows from one server install.

AI-native geospatial product builders

Founders and teams building product categories that didn't exist when the GIS stack was designed for human analysts at desktops — conversational geospatial intelligence, agent-driven site selection, AI-mediated remote sensing, autonomous mission planning. ForgeGIS is the compute substrate built for that thesis from the first commit.

PERFORMANCE HIGHLIGHTS

~800ms

warm-restart cold-start to first tools/list
response
returning all 62 MCP tools, list cached on first
build

14.46x-
39.75x

multi-stage GPU-resident pipelines vs.
GDAL CLI
measured warm at 2048²–4096² rasters

6.88x

geomean across 143 catalog ops with
GDAL CLI equivalent
May 2026 validation run, 127 wins of 143

Raw DE-9IM matrix for agent-encoded topology

Beyond the 8 OGC SFS named predicates (*Intersects, Disjoint, Within, Contains, Equals, Touches, Crosses, Overlaps*) and the 3 JTS extensions (*Covers, CoveredBy, ContainsProperly*), ForgeGIS exposes the raw 9-character DE-9IM intersection matrix per geometry-type pair through six matrix-computing operations. Agents reasoning about novel spatial relationships can encode topological queries that don't have named predicates — useful when an agent needs to express a relationship that the OGC vocabulary doesn't have a word for.

Build with ForgeGIS

A 21-page Technical Brief with the complete 220-operation catalog, MCP surface breakdown, validation methodology, and architectural deep-dive is available on request. Docker image and Maven coordinates available for evaluation.

rich@seaglassfoundry.com • seaglassfoundry.com